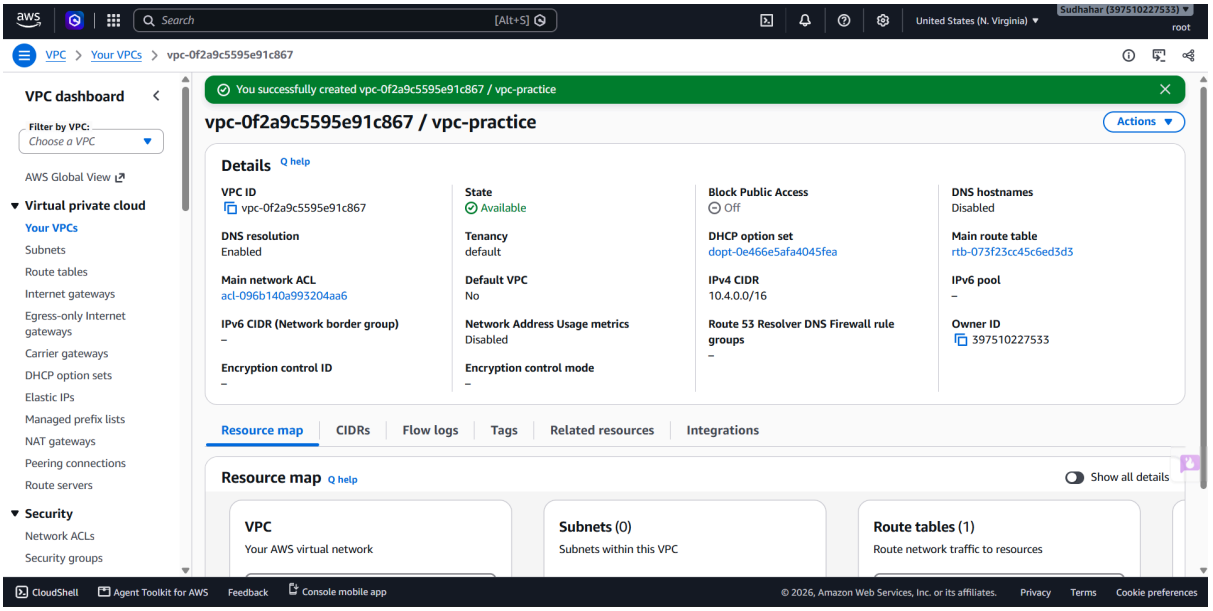
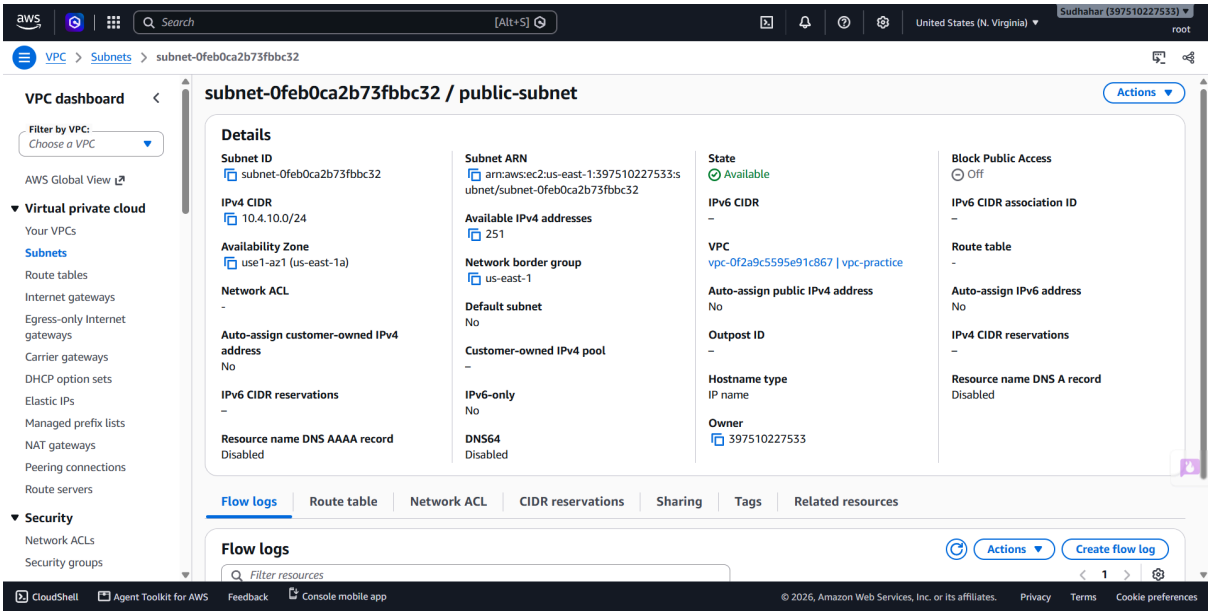
Date:02/11/2026

Task 3 - Working with VPC

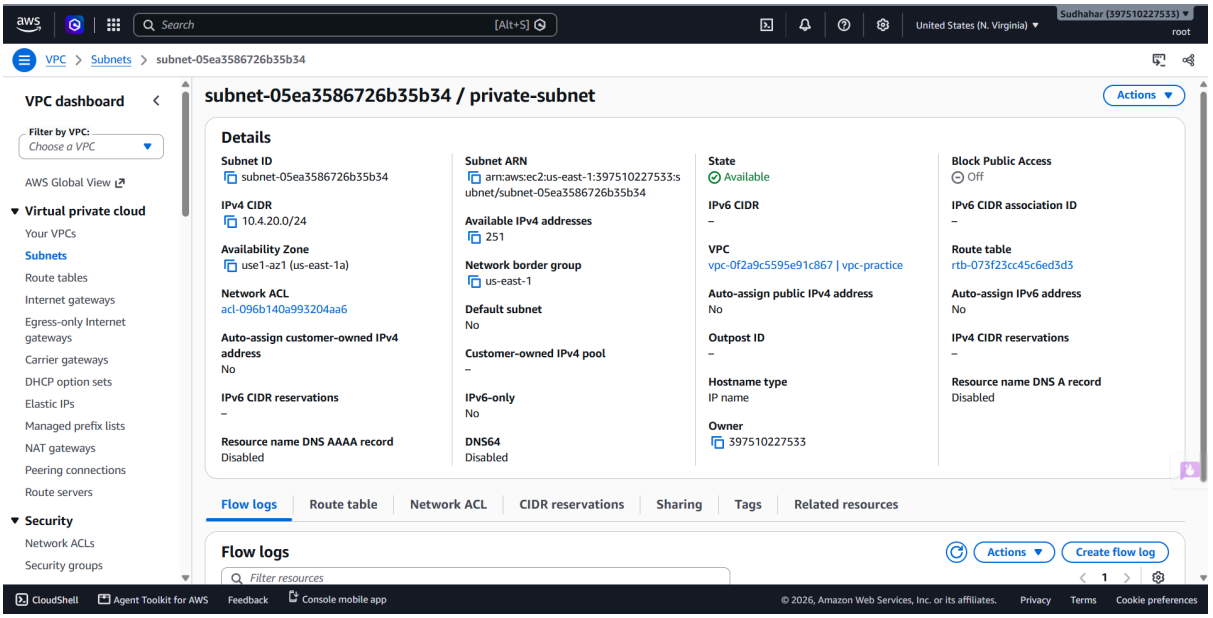
1.Create VPC:



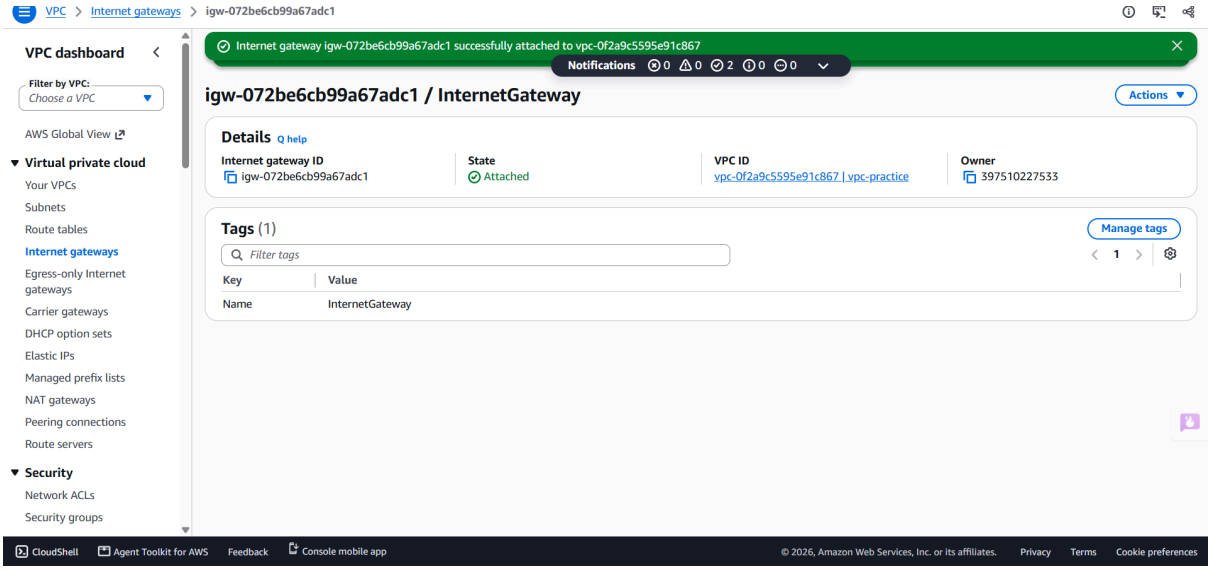
2.Create Public subnet:



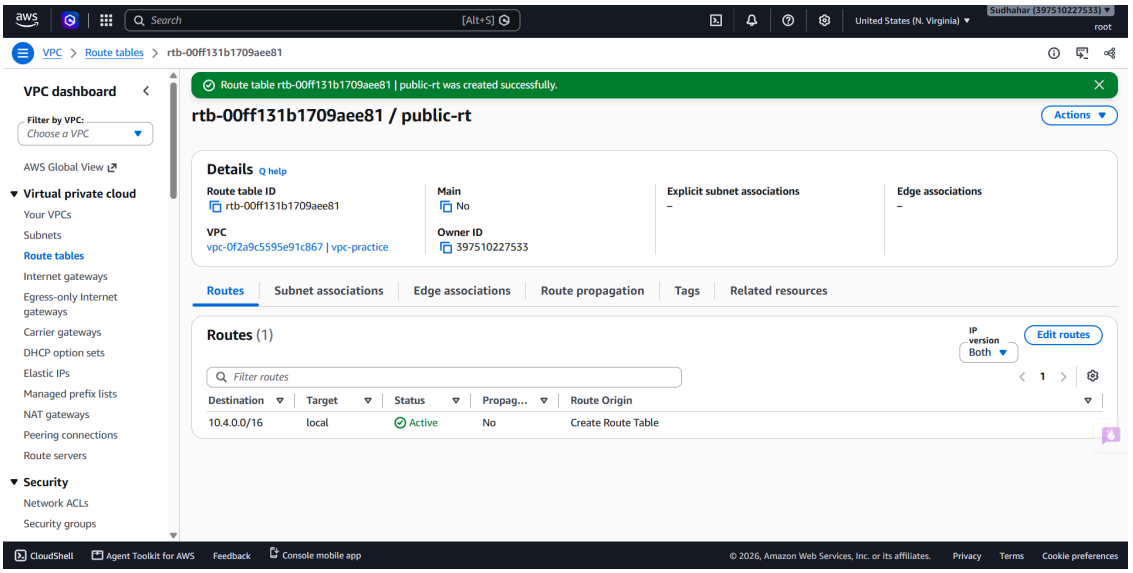
3.Private Subnet:



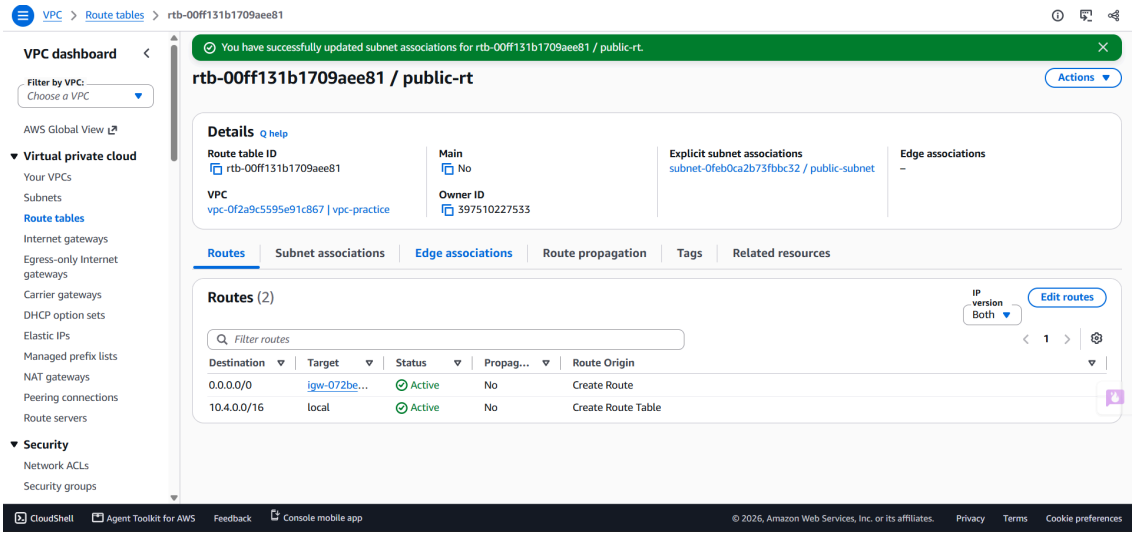
4.Create an Internet gateway:



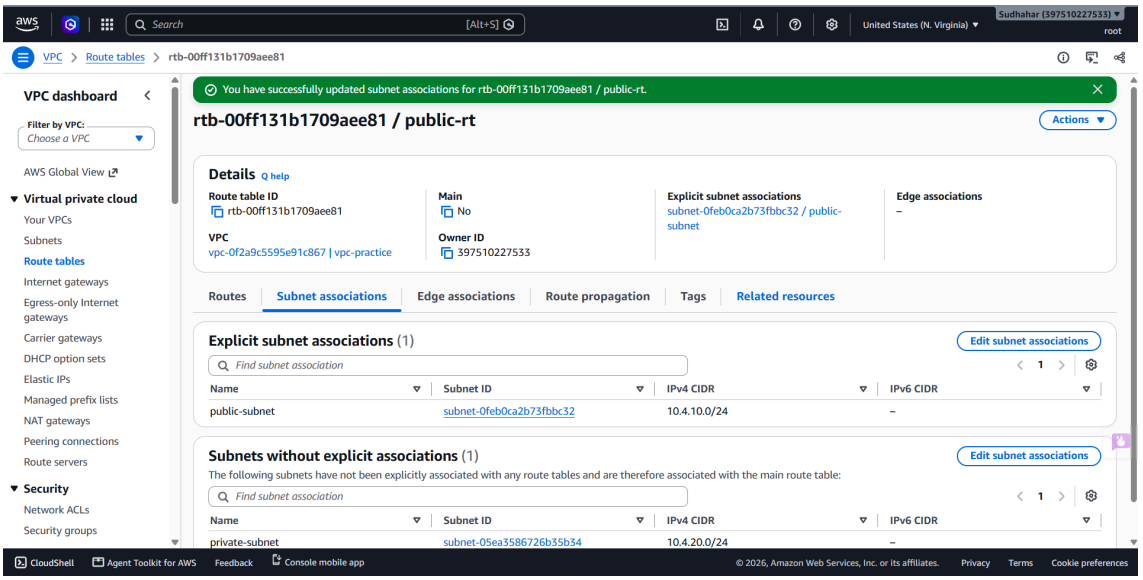
5. Create Public route table:



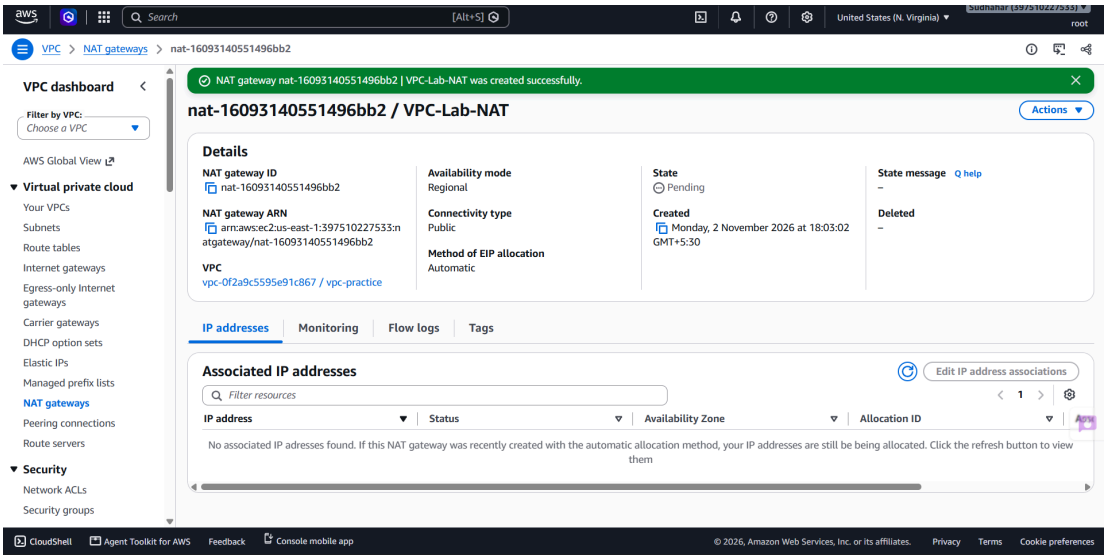
6.Edit Route for the public route table:



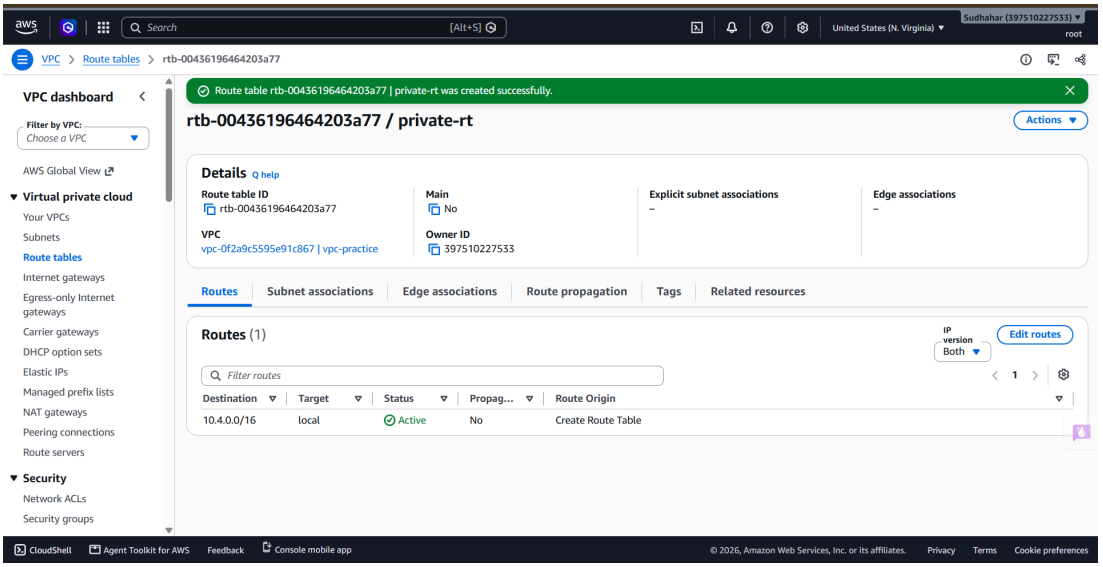
7.Do the subnet association:



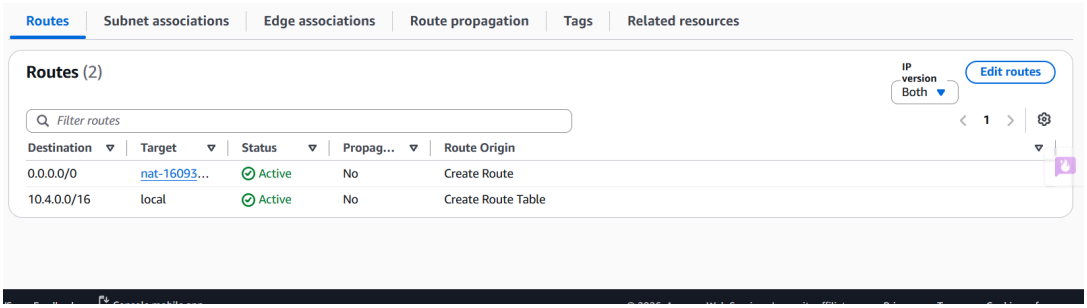
8.Create the NAT Gateway:



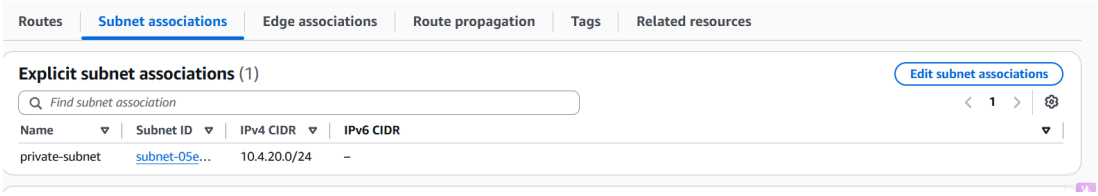
9.Create Private Route table:



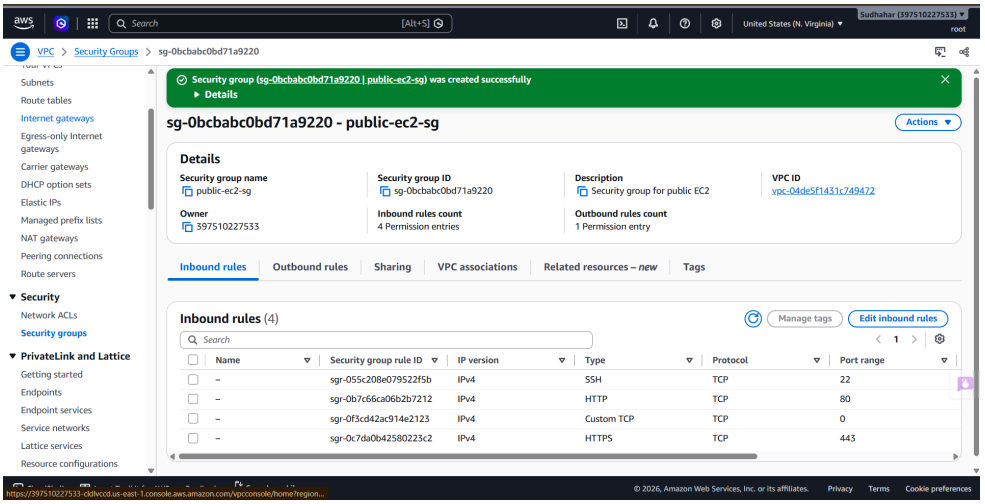
10.Edit route for the private route table:



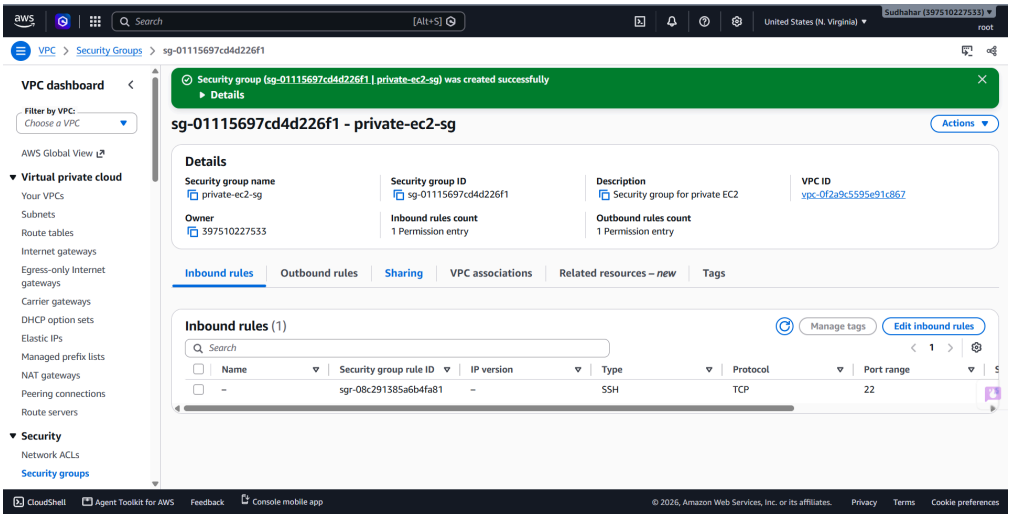
11.Do the subnet association for the private route table:



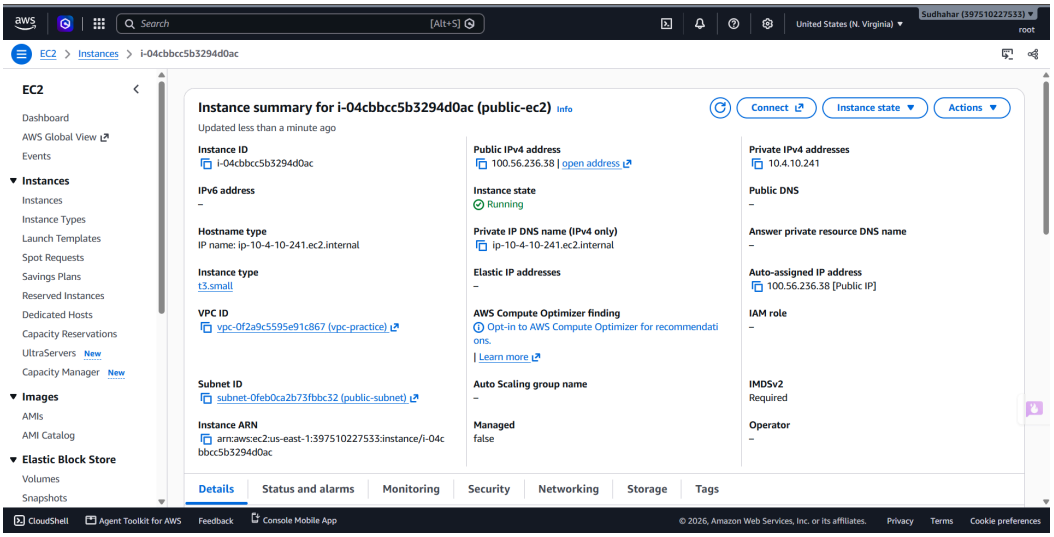
12.Create Security Groups: public security group



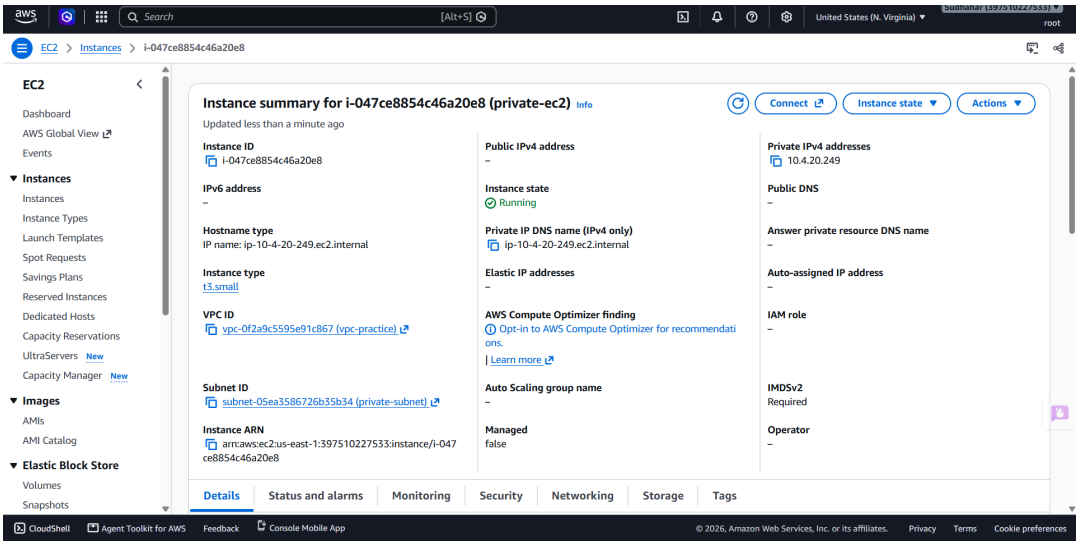
13.Private security group:



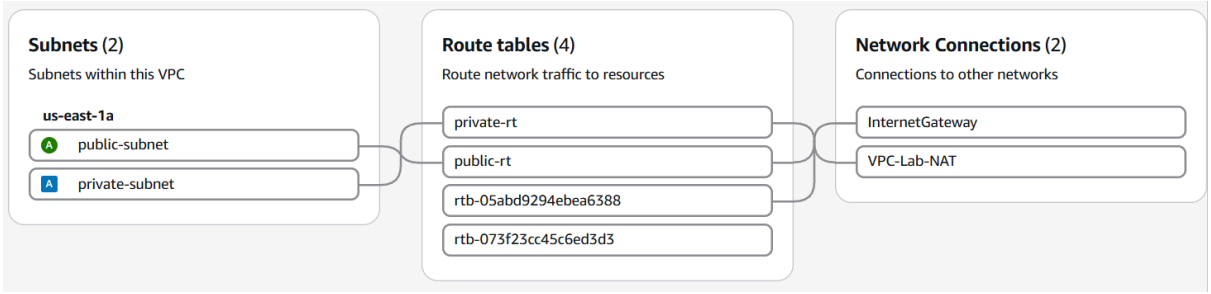
14.Public ec2 Instance:



15.Private EC2 Instance:

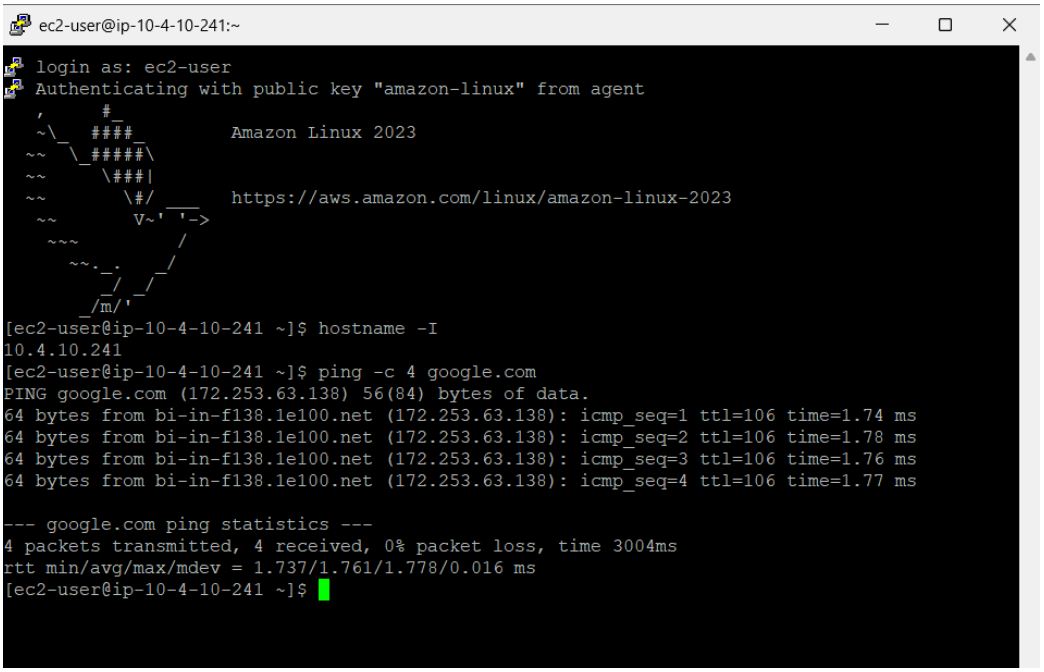


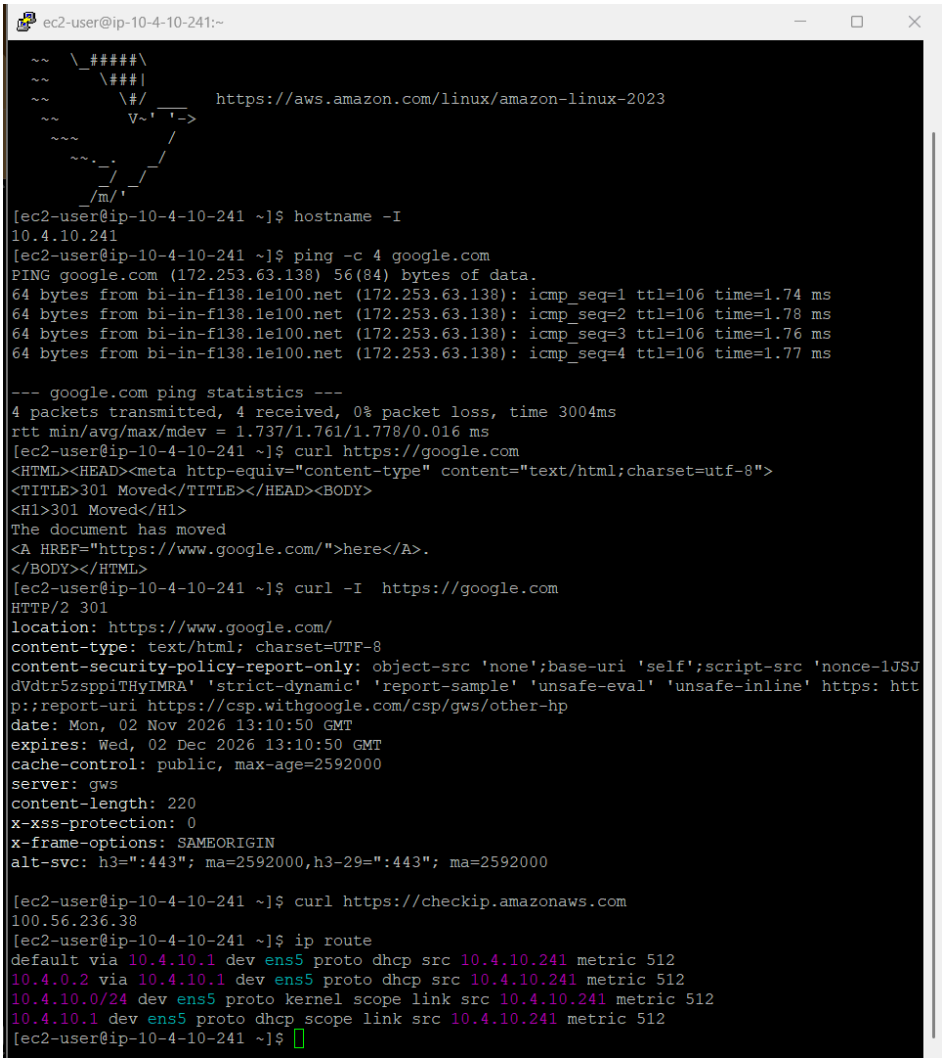
Final Resource map:



Output from terminal:

Public ip:





Private IP:

```
[ec2-user@ip-10-4-20-249 ~]$ hostname -I
10.4.20.249
[ec2-user@ip-10-4-20-249 ~]$ ip route
default via 10.4.20.1 dev ens5 proto dhcp src 10.4.20.249 metric 512
10.4.0.2 via 10.4.20.1 dev ens5 proto dhcp src 10.4.20.249 metric 512
10.4.20.0/24 dev ens5 proto kernel scope link src 10.4.20.249 metric 512
10.4.20.1 dev ens5 proto dhcp scope link src 10.4.20.249 metric 512
[ec2-user@ip-10-4-20-249 ~]$ curl -I https://example.com
HTTP/2 200
date: Mon, 02 Nov 2026 13:31:23 GMT
content-type: text/html; charset=utf-8
server: cloudflare
last-modified: Sun, 01 Nov 2026 22:45:37 GMT
allow: GET, HEAD
accept-ranges: bytes
age: 13371
cf-cache-status: HIT
cf-ray: a5437fcfdf344a43-IAD
alt-svc: h3=":443"; ma=86400

[ec2-user@ip-10-4-20-249 ~]$
```

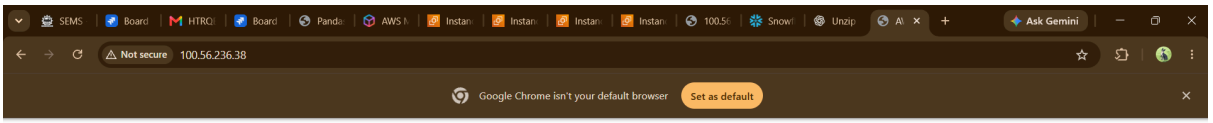
Index.html output:

```
[ec2-user@ip-10-4-10-241 ~]$ sudo nano /var/www/html/index.html
[ec2-user@ip-10-4-10-241 ~]$ ls -l /var/www/html/
total 4
-rw-r--r--. 1 root root 320 Nov  2 18:33 index.html
[ec2-user@ip-10-4-10-241 ~]$ curl http://localhost
<!DOCTYPE html>
<html>
<head>
    <title>AWS VPC Lab</title>
</head>
<body>
    <h1>Welcome to My AWS VPC Lab</h1>
    <h2>Public EC2 Web Server</h2>
    <p>This webpage is hosted on an EC2 instance in the Public Subnet.</p>
    <p>Public Subnet: 10.4.10.0/24</p>
    <p>Private Subnet: 10.4.20.0/24</p>
</body>
</html>
[ec2-user@ip-10-4-10-241 ~]$ ping -c 4 10.4.20.249
PING 10.4.20.249 (10.4.20.249) 56(84) bytes of data.

--- 10.4.20.249 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3132ms

[ec2-user@ip-10-4-10-241 ~]$ timeout 5 bash -c '</dev/tcp/10.4.20.249/22' && echo "Private EC
2 SSH reachable" || echo "Not reachable"
Private EC2 SSH reachable
[ec2-user@ip-10-4-10-241 ~]$
```

Output:



Welcome to My AWS VPC Lab

Public EC2 Web Server

This webpage is hosted on an EC2 instance in the Public Subnet.

Public Subnet: 10.4.10.0/24

Private Subnet: 10.4.20.0/24